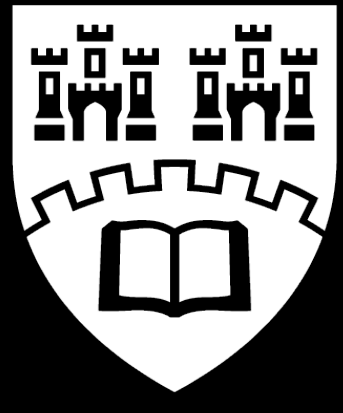




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A direct link between the solar magnetic configuration and Earth's radiation belts

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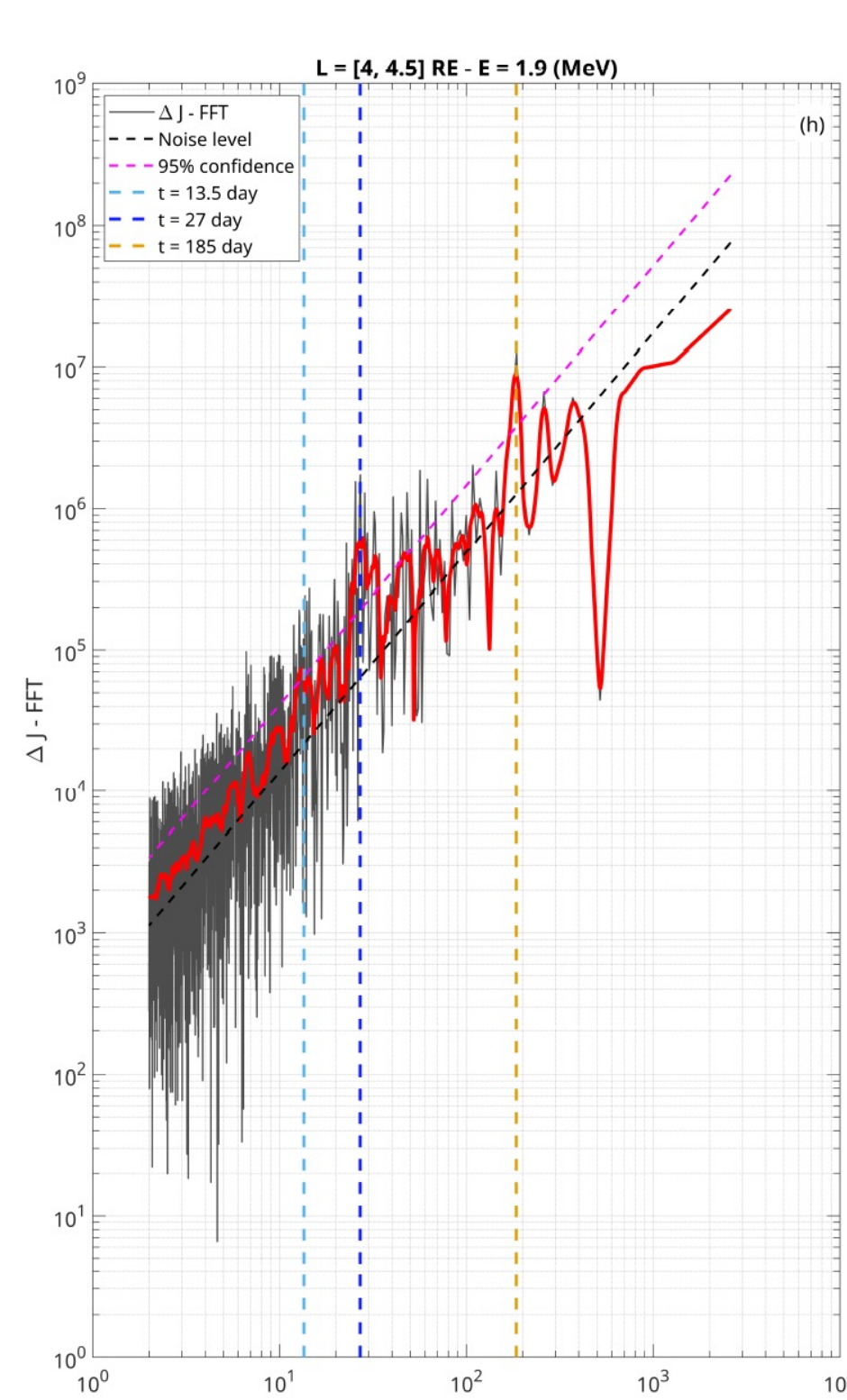
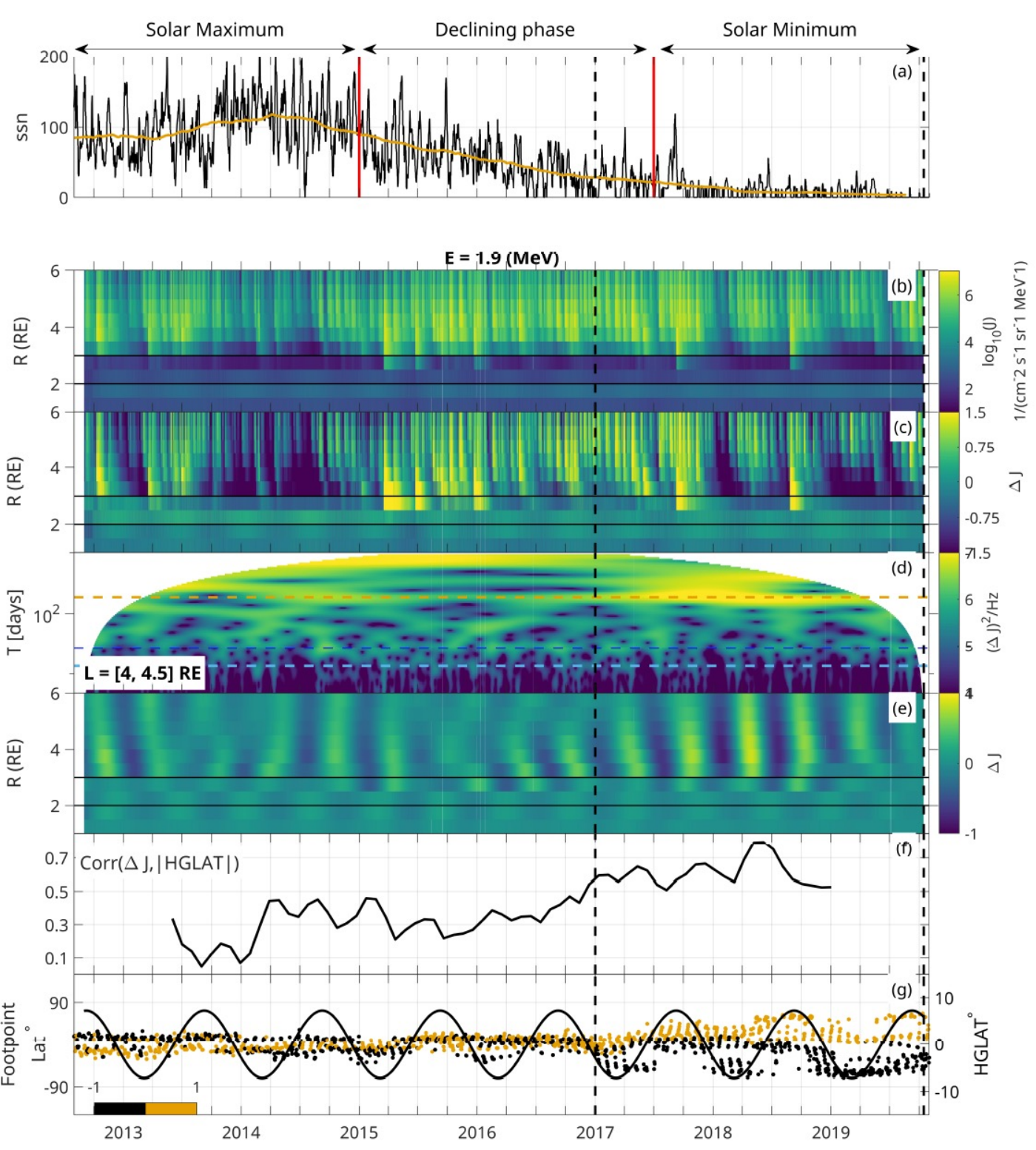
Introduction

The outer electron radiation belt exhibits periodic electron flux variations across multiple time scales – from solar cycle to seasonal and Carrington/sub-Carrington periods.

While solar cycle and Carrington variations are driven by the recurrence of high-speed solar wind streams, the cause of the seasonal modulation remains debated, often linked to geometric effects such as Russell-McPherron (RM) or equinoctial effect.

We show that variability in the solar magnetic configuration drives radiation belt electron flux variations across multiple periodicities, with periodic exposure to high-Alfvénicity solar wind enhancing the flux.

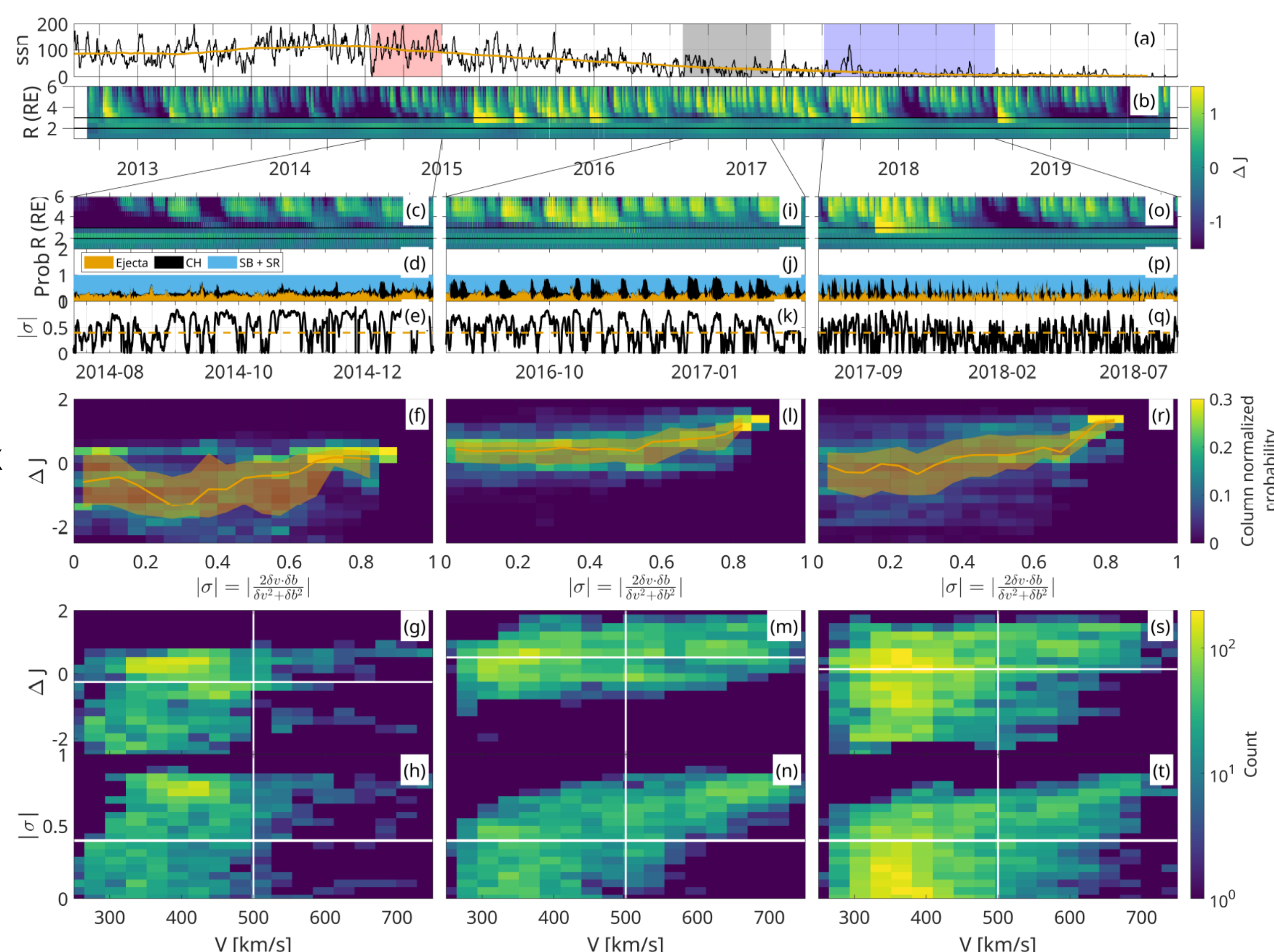
A revised mechanism for the seasonal variation of the radiation belt electron flux



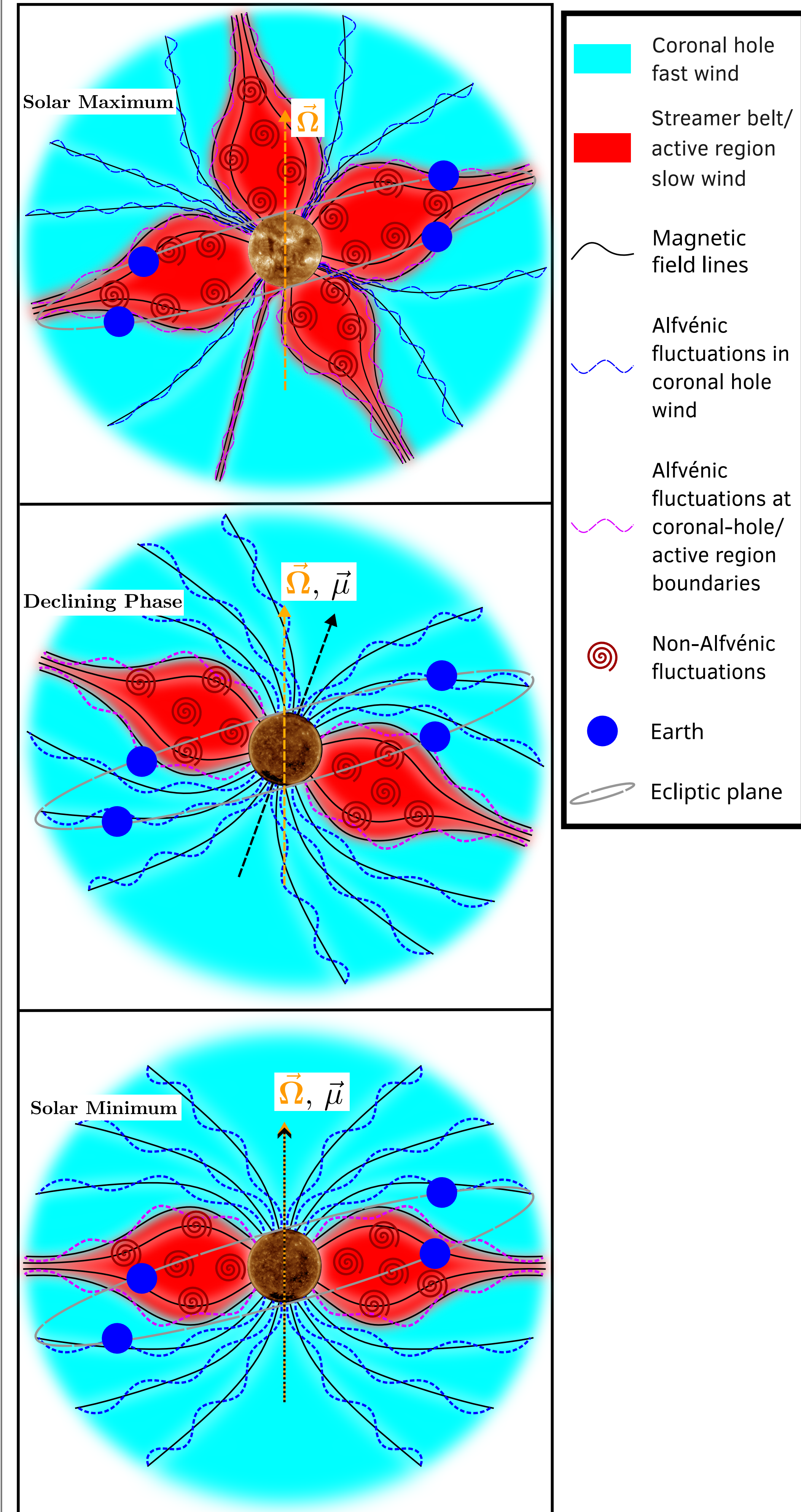
- Contrary to predictions of the RM and equinoctial effects, the seasonal variation peaks during solar minimum and nearly disappears during solar maximum
- During solar minimum, the longitude of the solar footpoint of the solar wind reaching Earth reaches polar latitude with alternating polarity. During other phases, its value is near equatorial.
- Electron flux varies with heliographic latitude (HGLAT), peaking about 30 days after the HGLAT maximum.
- **These observations are consistent with the axial effect as the primary driver of the seasonal variation in electron flux**

Influence of solar wind source and its cross helicity on radiation belt electron flux

- Recurrent electron flux enhancements occur at Carrington and sub-Carrington periods throughout the solar cycle.
- Throughout the solar cycle, flux enhancement is correlated with high Alfvénicity (σ) solar wind
- High speed streams can cause flux enhancement through two pathways:
 1. High σ --> enhanced day-side reconnection --> substorms --> flux enhancement.
 2. High solar wind speed --> Kelvin-Helmholtz instability --> ULF waves --> flux enhancement
- During solar maximum **slow-Alfvénic** wind causes flux enhancement
- **The above indicate that solar wind Alfvénicity rather than its speed is the main factor behind periodic radiation belt flux enhancement**



Summary and conclusions



- During solar maximum, the disappearance of the solar dipole results in a predominance of streamer belt solar wind at Earth, thereby reducing the overall electron flux.
- During the declining phase, the tilted solar dipole causes coronal hole solar wind to dominate at Earth, increasing overall electron flux.
- During solar minimum, the dipole moment is almost aligned with the solar rotation axis. This causes a seasonal variation of electron flux on top of the Carrington variation.
- The recurrence of high Alfvénicity solar wind (slow or fast) causes the periodic flux enhancement at the Carrington/sub-Carrington periods.